Line

Navigation for international professionals to avoid unforced errors on their NYC subway commutes

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TEAM

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line

Navigation for international professionals to avoid unforced errors on their NYC subway commute, from the first step underground to the right corner above it.

Team Next Steve Jobs
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1 THE PROBLEM

WHAT NEW YORKERS TOLD US

"People don't know where to go in the subway."

472	4.0M	46%
subway stations, many with several numbered exits	average weekday subway riders (MTA, 2025)	of NYC's workforce are immigrants (NYC Comptroller)

- **Wrong entrance** — Down the stairs to the wrong platform
- **Compass directions** — "Exit northwest" means nothing to a newcomer
- **Wrong exit** — Surface blocks away from the office
- **No signal** — Maps fail exactly where help is needed

2 OUR SOLUTION: THE LINE APP

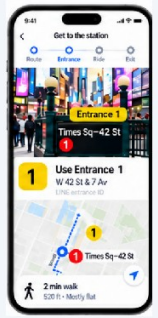
Every subway entrance and exit gets a number. Line maps **every step of the fastest route**, tells riders which door to board, **buzzes when they head the wrong way**, and names the exact exit for their destination. It works offline and in the rider's own language.

Plan the route



Fastest trip, door to door

Find the entrance



Right entrance, first try

Stand at the right door



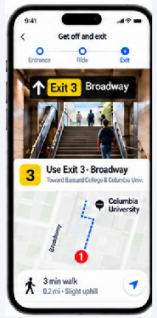
Board where your exit is

Wrong way? Feel a buzz



A nudge back on track

Take the right exit



Surface on the right corner

Know every exit



Street map, numbered exits

Follow your line in AR



Arrows in the real world

3 WHO IT'S FOR

International professionals moving to NYC for work
New city, new job, no time for wrong turns

- **MTA** — Fewer lost riders, less crowding at gates
- **Employers & relocation** — New hires arrive on time from day one
- **Families & visitors** — Same help for anyone new to a station
- **Businesses at exits** — Shown in "Around here" at each exit

4 HOW IT WORKS

Plan	Enter	Board	Buzz	Exit
Enter a destination; offline map downloads	Walk to the numbered entrance	Stand at the door nearest your exit	Wrong way? The phone vibrates	Take Exit 3, then turn left, not "go NW"

MTA open data: routes + exits · Indoor station maps · Works offline · Multilingual · Phone-camera AR

5 WHAT RIDERS SAID

> **Numbered exits matter most; she never knows where northeast is.**
> MOM ON THE MOVE · PARENT

> **Has taken the wrong entrance many times; the buzz would help most.**
> SEOUL SEARCHER · NEW FROM KOREA

> **Showing the right exits is what worked for him in Japan.**
> NEWBIE NAVIGATOR · NEW FROM CHILE

3 of 3 riders chose Line as the most helpful option over Exit Strategy and the MTA app

6 ECOSYSTEM

- **MTA data** · routes, exits
- **Station signs** · exit numbers
- **Employers** · new hires

line — app + offline station maps

- **Riders** · phone
- **Wearables** · buzz
- **Glasses** · AR arrows

7 LINE VS. ALTERNATIVES

	Line	Exit Strategy	MTA app	Google Maps
Names the exact exit	✓	✓	×	×
Guides inside the station	✓	Partial	×	×
Buzzes when you go wrong	✓	×	×	×
Left / right, not compass	✓	×	×	×
Works with no signal	✓	✓	×	×
AR arrows in the station	✓	×	×	×

8 AR & WEARABLES

- **AR on the phone camera** — Arrows on the station floor, no extra hardware
- **Any wearable** — Watch buzz and smart-glasses arrows
- **Underground maps** — Station interiors, platform to street

From your first step into the subway until you're back up in the city, Line finds your way.

Sources: MTA ridership 2025 · NYC Comptroller · team interviews and rider feedback, NYC, Oct 2026

CUSTOMERS | TARGET USERS & STAKEHOLDERS

Our user is the international professional new to NYC; the MTA, employers and station businesses gain too

PRIMARY USER

International professionals moving to New York for work

46% of New York City's workforce are immigrants (NYC Comptroller)

Needs: reach a new office on time from week one

Faces: unfamiliar stations, compass-style directions, no signal underground

Rides: a system of 472 stations and 4.0M weekday riders (MTA, 2025)

Families & visitors

Secondary users. Anyone new to a station gets the same help, in their own language.

MTA

Fewer lost riders and crowding at gates; data on where riders go wrong.

Employers

New international hires arrive on time and settle in faster.

Businesses at exits

Listed under "Around here" at each numbered exit, bringing foot traffic.

Wearable makers

A daily use case: buzz on the wrist, arrows in the lens.

Station agents

Fewer "which exit?" questions at the booth.

RESEARCH | INSIGHTS

Riders don’t need more maps; they need certainty at every decision point underground

	INSIGHT	WHAT NEW YORKERS TOLD US	SO LINE…
01	**Compass directions fail newcomers**	Mom on the Move never knows where northeast is and wants left/right turns after exiting, not north/south.	**Line** numbers every exit and says “turn left,” never “go NW.”
02	**The first mistake happens at the entrance**	Seoul Searcher has taken the wrong entrance many times. Interviewees struggle to find the right entrance on the first try.	**Line** routes riders to the numbered entrance for their platform.
03	**Newcomers already trust numbered exits**	Newbie Navigator used maps showing the correct exits in Japan and wants the same approach in New York.	**Line** brings a system riders know from Tokyo and Seoul to NYC.
04	**Small errors compound into misery**	“Traffic × Problems = Misery.” Delays are out of riders’ control; wrong entrances and exits pile avoidable errors on top.	**Line** removes the avoidable errors, so a delay stays a delay.
05	**Guidance must live on the rider’s own device**	Riders won’t share wearables for hygiene reasons, and many already wear glasses; a phone-camera version reaches everyone.	**Line** runs on the phone and pairs with riders’ own wearables.

Source: team interviews with NYC commuters; concept feedback from three NYC riders (Oct 2026)

Subway users need a way to avoid unforced errors on their commute

PROBLEM STATEMENT

“People don’t know where to go in the subway”



For

International people moving to the city for work

Who need to

Get around New York

Design

A way to avoid unforced errors on their commute

SOLUTION | IDEAL EXPERIENCE

Ideally, a new hire gets from front door to office without one wrong turn or moment of doubt



	Plan	Enter	Board	Ride	Exit & arrive
Today *Feels: anxious, late, embarrassed*	Copies a route, then guesses once inside the station	Takes the wrong entrance and crosses the avenue to try again	Picks the wrong direction, boards at the far end of the train	No signal; unsure if this is the right train or stop	“Exit NW” — surfaces on the wrong corner, arrives late
With Line *Feels: calm, on time, like a local*	One route, door to door, downloaded for offline use	Walks straight to Entrance 1 on the first try	Waits at Front car · Door 1, right where the exit will be	Phone buzzes the moment the direction is wrong	Takes Exit 3, turns left, walks in on time

Experience goal: zero unforced errors, from the first step underground to the right corner above it

SOLUTION | HOW IT WORKS

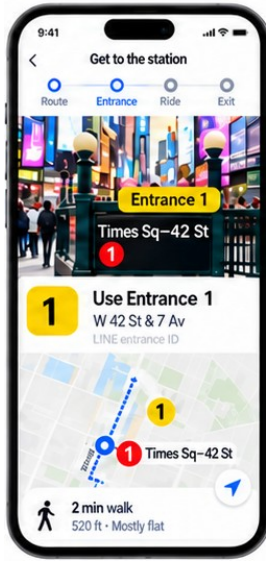
Line guides riders through every checkpoint, catches mistakes as they happen and can project the route in AR

1 Plan



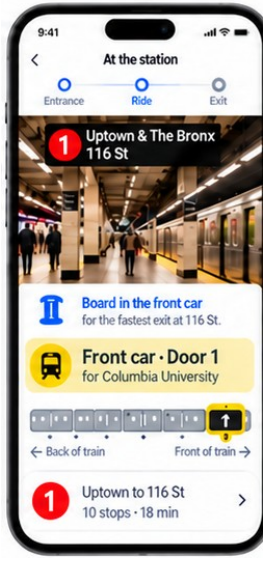
Enter a destination; the route and station maps download for offline use.

2 Enter



Line names the numbered entrance for the right platform.

3 Board



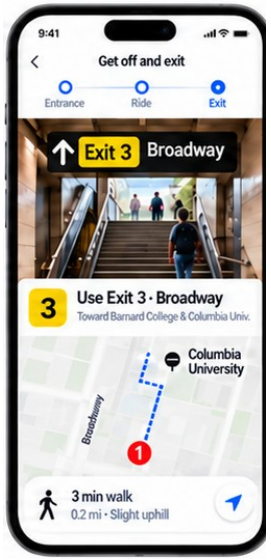
Shows the car and door that stop nearest the exit.

4 Buzz



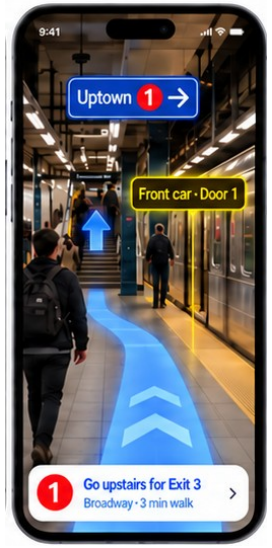
Wrong direction? The phone vibrates and points the way back.

5 Exit



Names the exit, then gives left/right turns to the door.

6 AR view



Hold up the phone or wear smart glasses: arrows appear on the station floor.

UNDER THE HOOD

MTA open data
Schedules, live trains, every entrance and exit

Indoor station maps
Platforms, stairs and doors

Phone sensors
GPS above ground; motion sensors below

Camera AR
Arrows on the floor via phone camera or smart glasses

Offline first
Every trip works with no signal

Rider's language
Directions in the rider's own language

SOLUTION | ECOSYSTEM

Line sits between the MTA's data and the rider's devices, and every partner gets something back

MTA

Gives Schedules, live trains, entrance and exit data

Gets Fewer lost riders; wrong-way hotspots to fix signage

Employers & relocation firms

Gives Line in the onboarding pack for new hires

Gets International hires on time from day one

Businesses near exits

Gives Sponsored "Around here" listings

Gets Foot traffic from riders leaving each exit

line

app + offline station maps

Riders

Gives Trips and feedback on where they got lost

Gets A commute with no unforced errors

Wearables & smart glasses

Gives Wrist buzz and in-lens AR arrows

Gets An everyday reason to wear the device

Phone platforms

Gives Camera AR, motion sensors, offline storage

Gets A flagship indoor-navigation app

MARKET | COMPARISON TO ALTERNATIVES

Line is the only option that guides riders inside the station and catches mistakes as they happen

	Line	Exit Strategy	MTA app	Google Maps	Asking a stranger
Names the exact exit to take	✓	✓	×	×	Partial
Shows where to stand on the platform	✓	✓	×	×	×
Guides you inside the station	✓	Partial	×	×	Partial
Buzzes when you go the wrong way	✓	×	×	×	×
Left / right, not compass directions	✓	×	×	Partial	✓
Works with no signal	✓	✓	×	×	✓
AR arrows	✓	×	×	Street only	×
Bottom line	Riders' #1 pick	Riders ranked 4–5	Riders ranked 4–5	Stops at the entrance	Not always there

Source: team review of each product and rankings from three NYC riders (Oct 2026). Exit Strategy costs $4; the MTA app and Google Maps are free.

ELEVATOR PITCH

From your first step into the subway until you’re back up in the city, Line finds your way

New York draws thousands of international professionals every year. They arrive ready for the job, then they reach the subway, where finding the right entrance, platform and exit is like navigating a jungle underneath the city.

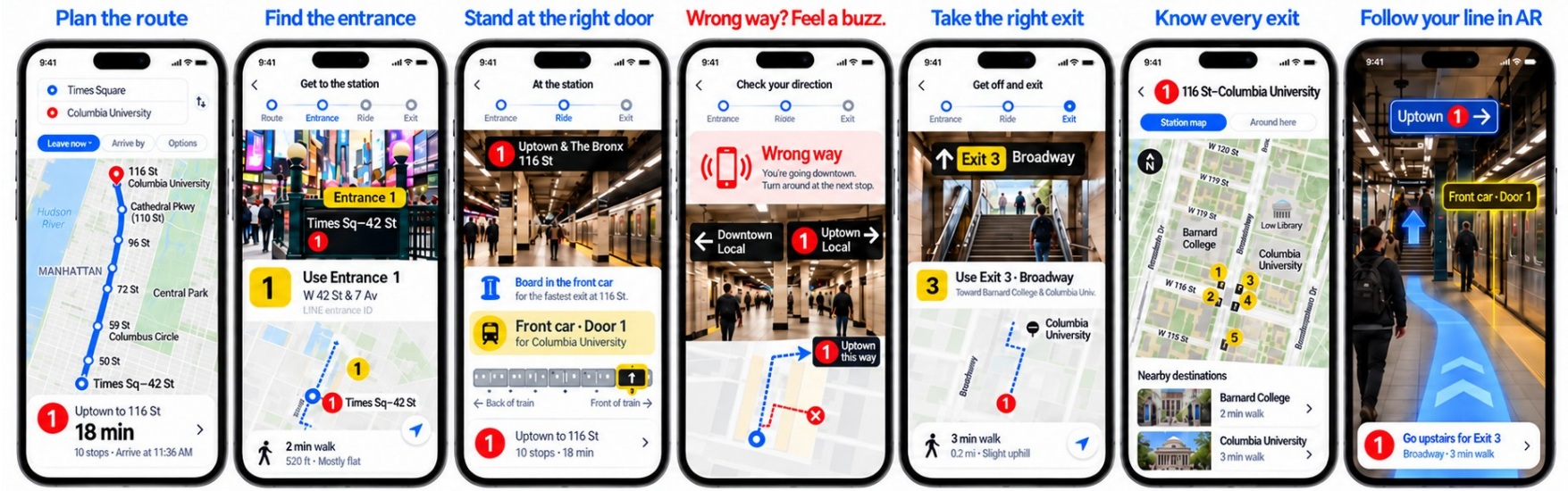
What they need is Line. With every subway exit numbered, the Line app maps each step of the fastest route, shows where to stand and buzzes the moment you head the wrong way, even with no signal. Line also projects augmented reality arrows through your phone or smart glasses.

Line is how the international professional navigates New York with confidence.

96 words

PROTOTYPE

Our prototype covers the whole trip



SHAPED BY RIDER FEEDBACK

Left/right after exiting

Turn-by-turn on the street, never “head northwest.” From Mom on the Move.

Underground station maps

Interior maps of platforms and passages. From Seoul Searcher.

AR on riders’ own devices

Phone-camera AR and riders’ own smart glasses. From all three riders.

Offline and multilingual

Works with no signal, in the rider’s own language. From our interviews.